Single-Cell RNA-seq Analysis of COVID-19 PBMCs: From Preprocessing to Predictive Modeling

Malha Entezari **Amir Hossein Ziaeifar** **Qazal Mousavi**

1. Introduction

This report presents a comprehensive analysis of a single-cell RNA sequencing (scRNA-seq) dataset derived from Peripheral Blood Mononuclear Cells (PBMCs) of patients with varying severities of COVID-19. The primary objective of this project was to establish a complete analytical pipeline, from raw data preprocessing to the development of predictive machine learning models, with a keen focus on biological interpretation and methodological rigor.

The dataset chosen for this study (GSE293707) comprises pooled samples from four distinct clinical categories: Healthy controls, and individuals with Mild, Severe, and Critical COVID-19. The analysis was structured in two main phases. The first phase involved a standard scRNA-seq preprocessing workflow, including quality control, normalization, dimensionality reduction, clustering, and cell type annotation. The second phase leveraged these results to build and evaluate machine learning models capable of predicting disease severity based on cell-type-specific gene expression profiles.

A key innovation of this project was the implementation of a Leave-One-Severity-Out (LOSO) cross-validation strategy. This approach was specifically designed to address the risk of data leakage, a common pitfall when dealing with multi-sample datasets, by ensuring that all cells from a single severity class were held out during model training. A key practical outcome of the project was demonstrating that data structure and experimental design strongly influence both statistical validity and machine learning performance. In particular, the lack of biological replicates imposed important limitations on classical statistical inference and required careful interpretation of results.

2. Data and Preprocessing

1. Dataset:

The dataset **GSE293707** consists of pooled PBMC (Peripheral Blood Mononuclear Cell) samples collected from both healthy individuals and COVID-19 patients. This dataset contains transcriptomic data derived from single-cell RNA sequencing (scRNA-seq) with a focus on immune cells and their response to infection. The dataset includes 72,809 cells from across the four samples and 36,602 genes across all cells.

- **Samples**: 4 GSM samples, where each sample represents a pooled collection of cells.
    - **GSM8888963**: Healthy patient pool
    - **GSM8888964**: Mild disease patient pool
    - **GSM8888965**: Severe disease patient pool
    - **GSM8888966**: Critical disease patient pool

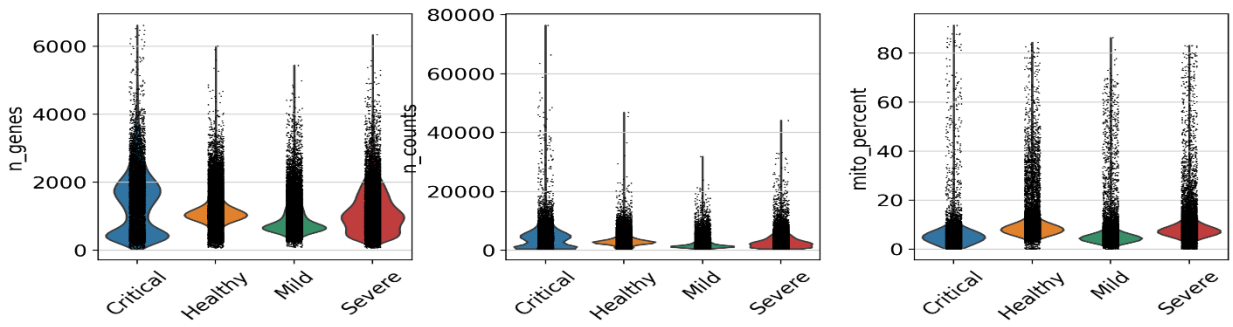
The dataset uses pooled samples, which means we are working with data from multiple individuals combined, rather than individual cell-level replication. This can obscure subtle differences between individual patients.

Another challenge identified throughout this project is the limited biological replication, specifically the presence of only one Healthy control sample against three disease samples. This imbalance introduces a high risk of pseudo-replication, where a machine learning model might inadvertently learn donor-specific genetic backgrounds rather than the general biology of the disease.

2.2 Preprocessing and Biological Analysis:

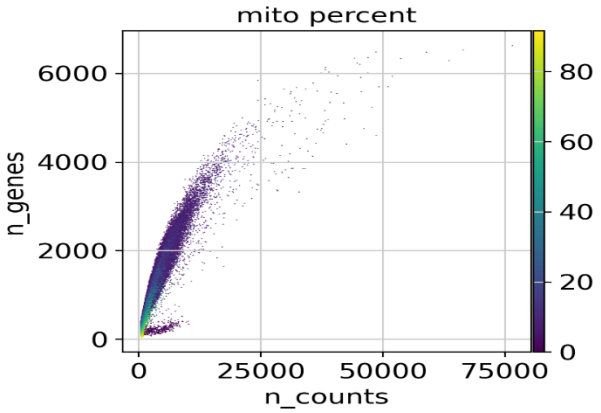
- **Quality Control (QC) and Filtering:**

The analysis began by loading the raw data and performing initial quality control. QC metrics were calculated, including the number of genes per cell (n_genes), the total number of UMI counts (n_counts), and the percentage of mitochondrial gene counts (mito_percent), a common indicator of cell stress or death.



The violin plots illustrate the distribution of these QC metrics across the four severity groups. While the distributions are broadly similar, there are observable differences, particularly in the upper ranges of n_genes and n_counts, which may reflect biological differences between the conditions.

The scatter plot shows the relationship between n_counts and n_genes, with mito_percent as a color gradient. This plot helps visualize the rationale behind the chosen thresholds, showing a cluster of cells with high mitochondrial content and low gene counts that were subsequently filtered out.



Based on these metrics, a filtering strategy was applied to remove low-quality cells and potential doublets:

- n_genes between 300 and 4000
- n_counts between 800 and 30,000
- mito_percent < 15%

After filtering, the dataset was reduced to 63,736 cells. Subsequently, genes detected in fewer than 3 cells were filtered out, resulting in a final expression matrix of 63,736 cells and 25,057 genes.

- **Normalization and Feature Selection:**

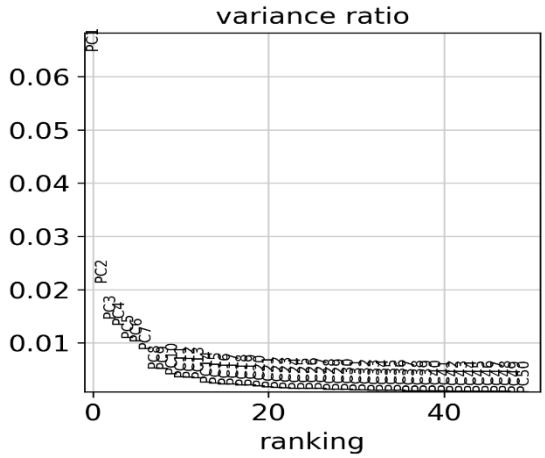
To account for differences in sequencing depth, the data was normalized using a target_sum=1e4, followed by a log transformation. It removes technical biases related to library size, allowing downstream analyses like clustering and differential expression to focus on the biological variation between cells, rather than differences in sequencing depth.

Highly variable genes (HVGs) were identified using the seurat_v3 method, with the gsm_id specified as a batch_key. This is a crucial step, as it ensures that the selected HVGs capture biological variation rather than technical or sample-specific differences. The top 3,000 HVGs were selected for downstream analysis. The data was then scaled to have unit variance and zero mean.

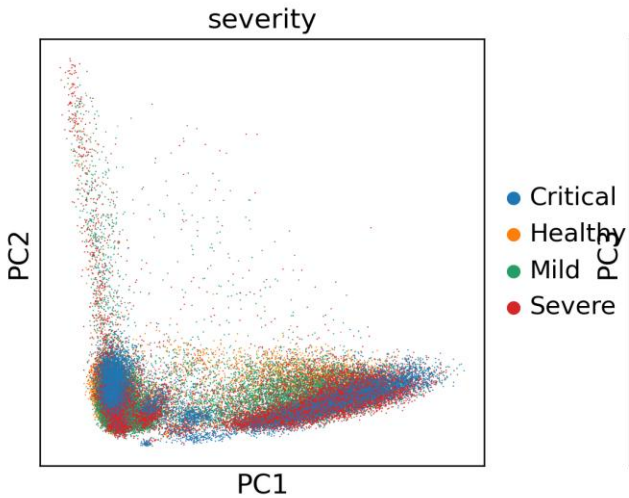
- **Dimensionality Reduction and Clustering:**

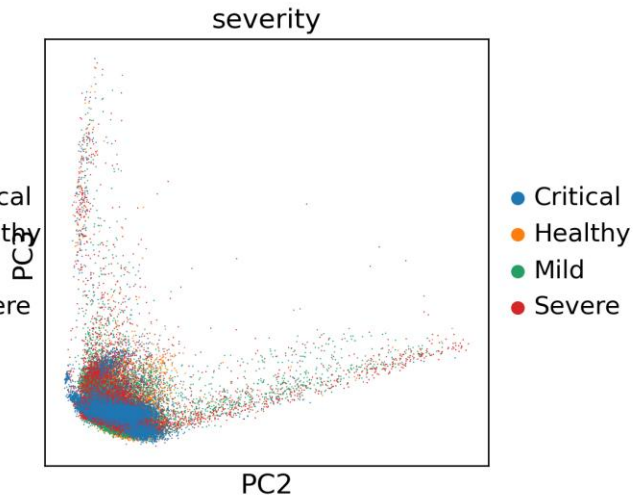
PCA is a common technique for reducing the number of dimensions in large datasets like scRNA-seq. PCA was applied using the top 3000 HVGs, reducing the dimensionality from thousands of genes to a smaller number of principal components that capture the most variation in the data.

The PCA variance ratio plot showed a sharp decrease in variance explained within the first 10 principal components, followed by a gradual plateau after approximately 15–20 PCs. Therefore, 20 principal components were selected to capture the major biological variation while minimizing noise.

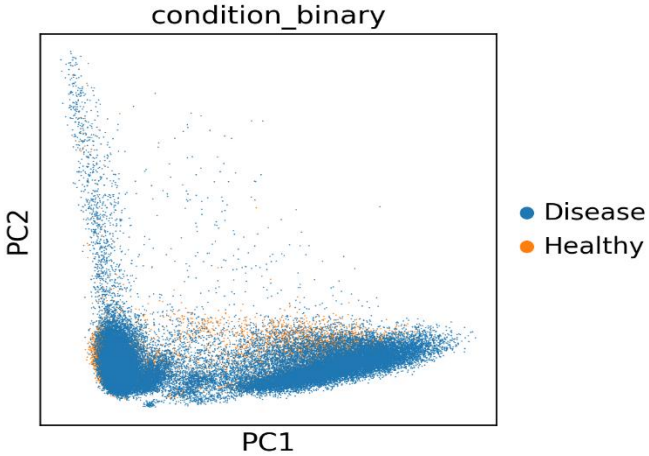


The PCA plots show that most variation is captured by PC1, with a typical L-shaped structure seen in heterogeneous PBMC data. When colored by severity, there is no complete separation between groups, but a gradual shift from Healthy to more severe conditions can be observed.

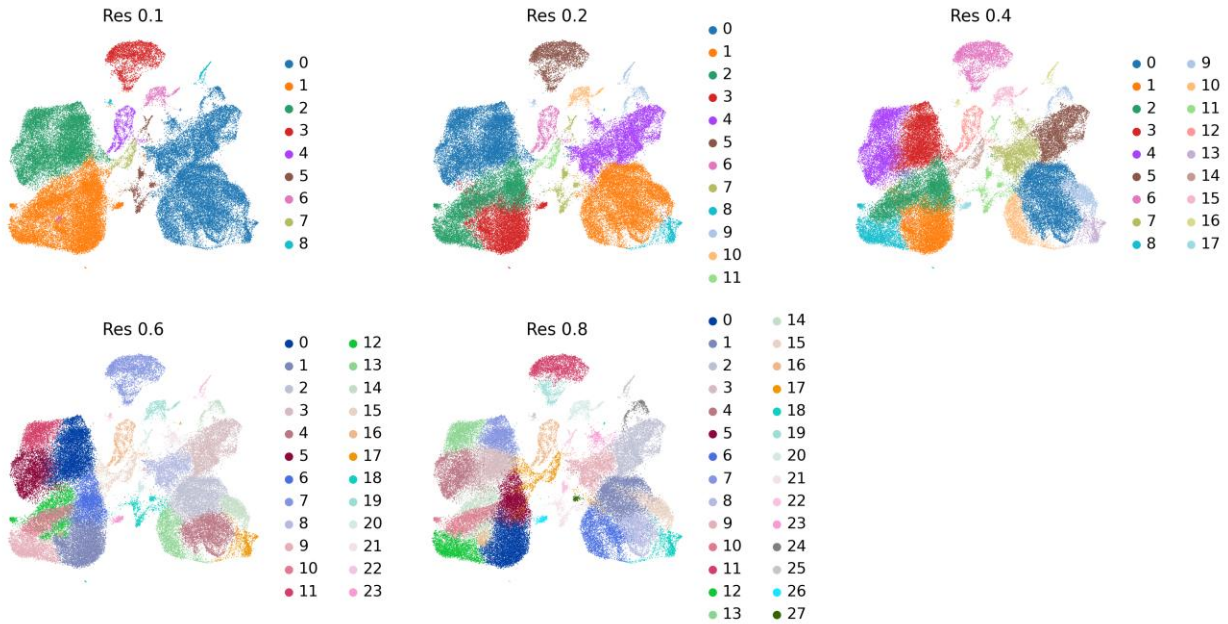




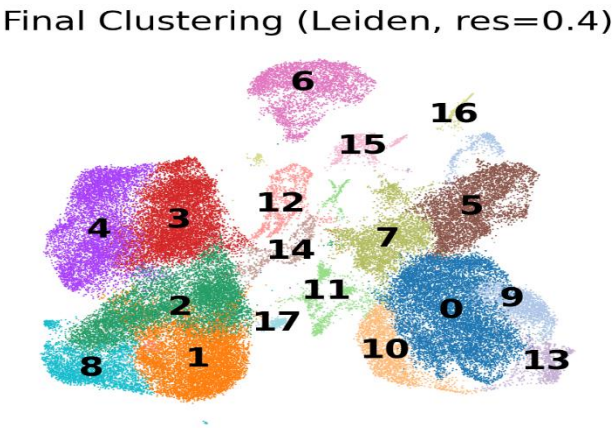
Overall, the binary condition (Healthy vs Disease) also shows substantial overlap, which is expected for an unsupervised method like PCA and reflects biological complexity rather than technical bias.



Clustering was performed using the Leiden algorithm on the PCA-reduced data. This method identifies distinct groups of cells (clusters) that share similar gene expression profiles. To determine the optimal clustering granularity, Leiden clustering was performed across multiple resolution parameters (0.1, 0.2, 0.4, 0.6, and 0.8). The objective was to identify a biologically meaningful partitioning of the PBMC dataset while avoiding both under-clustering (over-merging distinct populations) and over-clustering (artificial fragmentation of homogeneous populations).

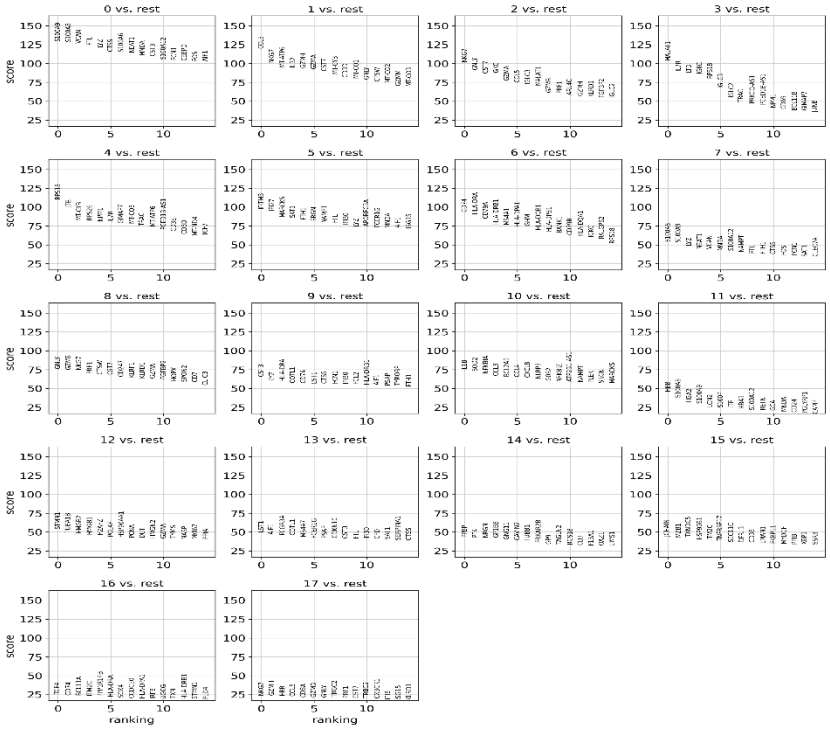


Resolution 0.4 provided the best balance between biological interpretability and cluster stability. Major cell populations were clearly separated while preserving structural coherence in the embedding space. Additionally, cluster sizes remained sufficiently large to support robust differential expression testing and supervised modeling.



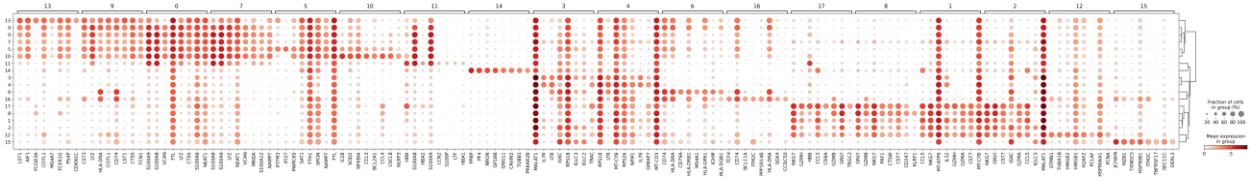
- **Cell Type Annotation and Differential Expression Analysis:**

Clusters were annotated based on the expression of canonical marker genes, as shown in the plot.



The following major cell types were identified:

- CD14+Monocytes: *S100A8, S100A9, LYZ*
- FCGR3A+Monocytes: *MS4A7, LST1*
- CD4+ T Cells: *CD3D, IL7R*
- CD8+ T Cells: *CD8A, GZMK*
- NK Cells: *GNLY, NKG7*
- B Cells: *MS4A1, CD79A*
- Plasma Cells: *MZB1, JCHAIN*
- Dendritic Cells (DC): *CLEC9A, FCER1A*
- Megakaryocytes: *PPBP*



The number of biological replicates in this dataset is very limited (only 4 GSMs). As a result, the cell type annotation process could be biased towards the cells most represented in the data. This means that some minor or rare cell types might not be well-represented or might not be identified accurately.

Differential expression analysis was performed globally to identify genes that define the transcriptional signature of each severity group. A critical limitation must be acknowledged: due to the dataset's structure—with only one biological sample per condition—formal statistical testing (e.g., using a pseudo-bulk approach) is not feasible. The observed differences are confounded by inter-sample variability, preventing us from drawing statistically robust conclusions about condition-specific gene expression changes. Results were used for feature selection in downstream machine learning rather than formal biological inference

- **Data Splitting Strategy for Machine Learning:**

For the machine learning phase, a Leave-One-Severity-Out (LOSO) cross-validation scheme was implemented. This approach was specifically designed to address the dataset's structure and prevent data leakage. In each fold of this validation scheme:

- Training Set: Consists of two complete disease severity groups (e.g., Mild and Severe) along with half of the Healthy control cells.
- Test Set: Consists of the remaining one disease severity group (e.g., Critical) along with the other half of the Healthy control cells.

This process is repeated for each severity group, resulting in three validation folds (with Healthy always split between training and test). By ensuring that cells from the same disease severity group are not shared between training and test sets, the model is forced to learn generalizable disease-associated signals rather than severity-specific patterns.

However, due to the structure of the dataset, all Healthy cells originate from a single biological sample. As a result, splitting Healthy cells between training and test sets introduces a potential source of within-sample data leakage. In this scenario, the model may partially learn donor-specific or technical patterns rather than purely biological differences.

Therefore, while the LOSO strategy effectively prevents leakage across disease severity groups, it does not completely eliminate all forms of data leakage. The results of the machine learning models should be interpreted with this limitation in mind.

3. Analysis

In this section, we will review the different models used for classification, analyze the results of each approach, and compare the performance of different models.

3.1 Models:

In this section, we will review the different models used for classification, analyze the results of each approach, and compare the performance of different models.

For the predictive modeling task, we evaluated four different classifiers on their ability to predict the four severity classes based on the average gene expression profiles of each cell type for each sample. The models used were:

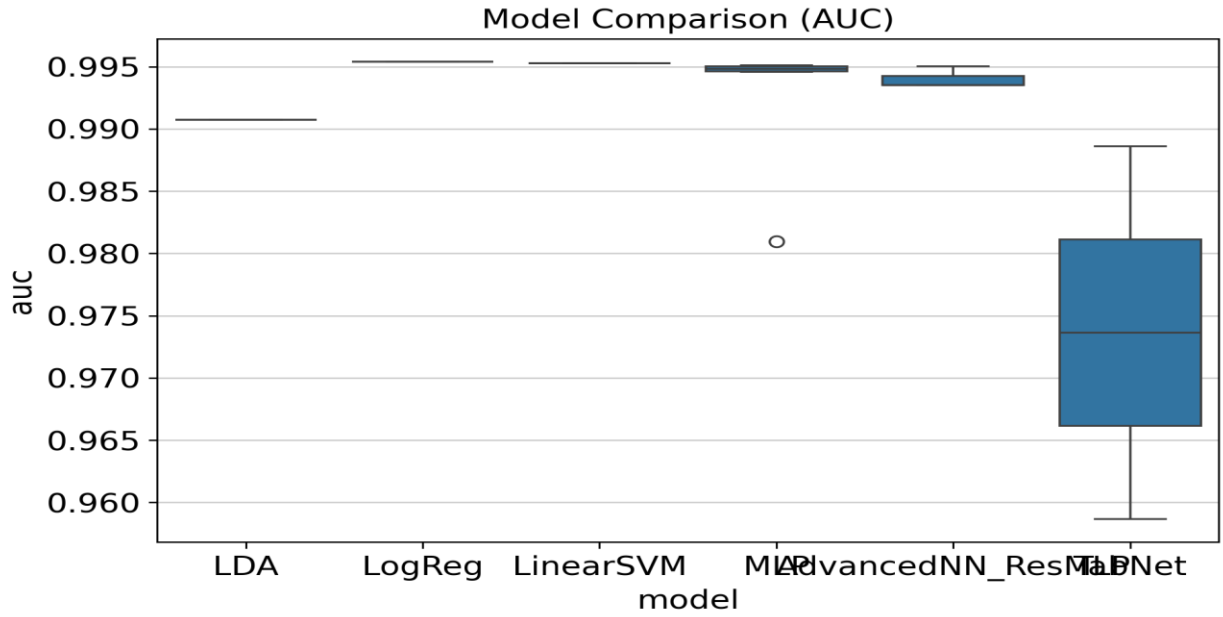
1. **Linear Discriminant Analysis (LDA)**: A generative model that finds a linear combination of features that best separates two or more classes.

2. **Logistic Regression (LogReg)**: A discriminative model that uses a logistic function to model the probabilities of class membership.
3. **Support Vector Machine (SVM) with a Linear Kernel**: A discriminative classifier that finds the optimal hyperplane to separate classes in the feature space.
4. **Advanced Neural Network**: A multi-layer perceptron model with the capacity to learn complex, non-linear relationships in the data.

3.2 Result Analysis:

- **Approach 1(Flawed / Leakage-Affected Analysis):**

In the initial modeling approach, the results appeared exceptionally strong across all models, with Area Under the Curve (AUC) values consistently exceeding 0.95. At first glance, such performance might suggest that the selected gene features are highly discriminative and that the classification task is trivial. However, upon closer examination, this outcome is not biologically or statistically plausible, especially given the known variability and noise inherent in single-cell RNA-seq data.



The root cause of this issue lies in data leakage during feature selection. Specifically, the model used Differentially Expressed Genes (DEGs) derived from Phase 7 exploratory analysis, where gene selection was performed using the entire dataset. This means that both training and test cells contributed to identifying the most discriminative genes. As a result, the model indirectly gained access to information from the test set before evaluation.

This leakage creates an artificially favorable scenario in which the features are already optimized to separate the exact data the model is later tested on. Consequently, the model is not truly learning generalizable biological patterns but rather exploiting pre-exposed information. This explains why:

- All models (LDA, Logistic Regression, Linear SVM) achieved near-perfect performance
- Performance remained consistently high across all cell types and severity splits
- There was little to no variation between models

From a machine learning perspective, such uniformity is a strong indicator that the evaluation process is compromised. In realistic biological classification problems—especially those involving pooled samples and limited biological replication—performance should vary across cell types and disease severities.

Another contributing issue in this phase was the lack of strict control over data splitting consistency. In some cases, models were effectively evaluated on slightly different subsets of data, making comparisons between models less reliable. This further reduces the interpretability of the results.

In summary, although the first approach produced impressive numerical results, these results are not trustworthy. The presence of data leakage invalidates the evaluation, and the conclusions drawn from this phase cannot be considered biologically meaningful.

- **Second Approach (Corrected and More Reliable Analysis):**

To address the limitations of the first approach, a revised modeling strategy was implemented with a strong emphasis on reducing data leakage and ensuring fair evaluation.

The most important correction was enforcing a strict separation between training and test data. For each cell type, healthy cells were randomly split once into two disjoint subsets (50% for training and 50% for testing), ensuring that no healthy cell appeared in both sets. Disease cells were then partitioned using a leave-one-severity-out strategy, where one severity group (e.g., Mild, Severe, or Critical) was used exclusively for testing while the others were used for training.

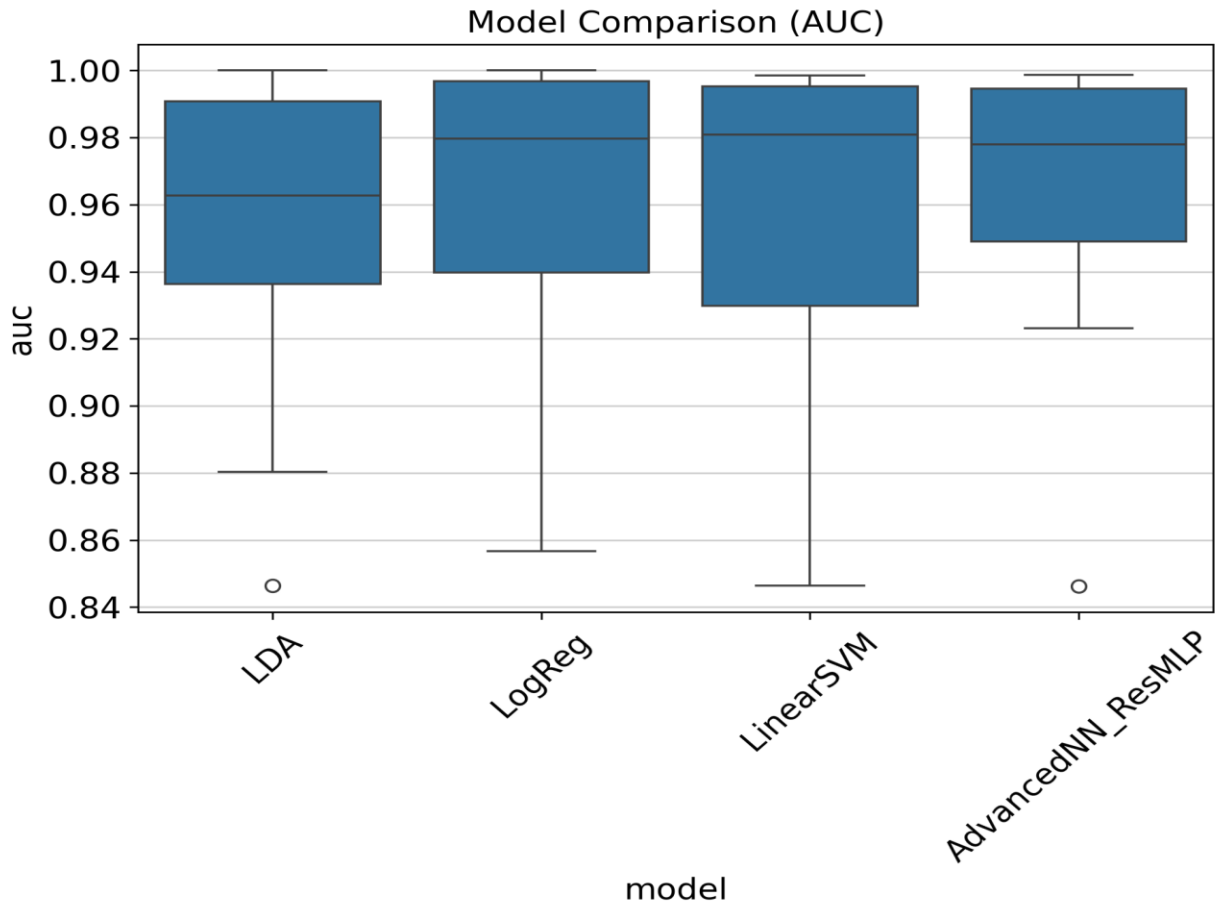
This design has two important consequences:

It eliminates overlap between training and test sets at the cell level. It evaluates model generalization across disease severities, which is biologically meaningful

Another key improvement was ensuring that all models were trained and evaluated on exactly the same data splits. This eliminated inconsistencies present in the first approach and enabled fair comparison across models.

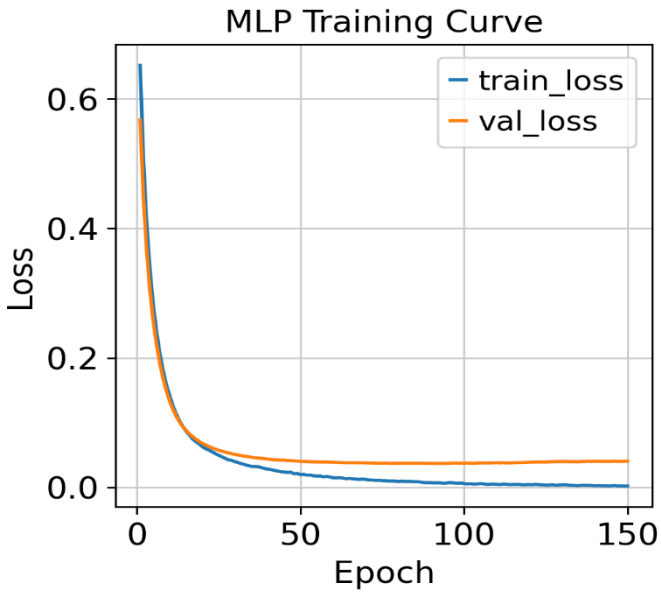
Although DEG genes from Phase 7 were still used as input features, it is important to note that these genes were originally identified using the full dataset. Therefore, a mild form of feature-level information leakage may still be present. However, since the data splitting was strictly enforced before model training and evaluation, the impact of this leakage is significantly reduced compared to the initial approach.

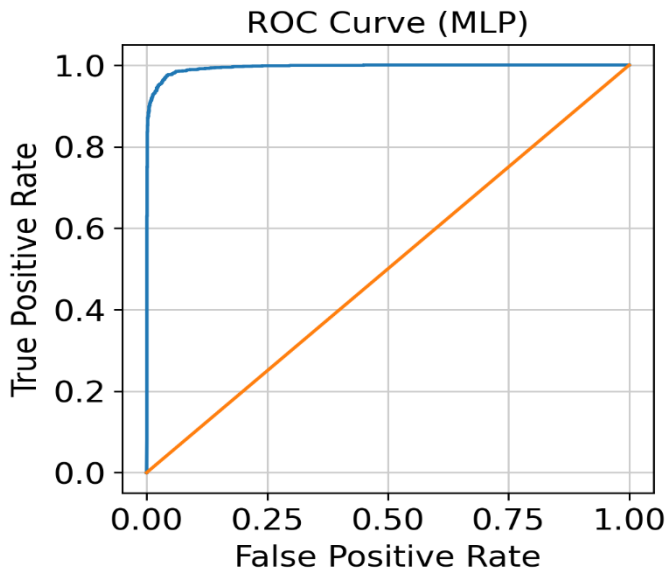
After applying these corrections, model performance dropped to a more realistic range compared to the initial approach. This reduction is expected and reflects a more honest estimate of the model's predictive capability.



In addition to the classical machine learning models, a neural network–based model (MLP) was trained using a more rigorous training procedure that included an internal validation split and monitoring of training dynamics. This model was implemented through a dedicated function that splits the training data into training and validation subsets using stratified sampling, ensuring class balance during optimization.

The MLP achieved MLP Test Accuracy 0.964 and AUC 0.995. The MLP Training Curve plot shows rapid early convergence, with train_loss 0.6522 and val_loss 0.5680 at Epoch 01/150, followed by sustained improvement that drives train_loss to 0.0020 at Epoch 150/150 while val_loss is 0.0404 at Epoch 150/150. The widening train–validation gap in later epochs is consistent with overfitting in loss space, even though the final discrimination performance remains very high. The ROC Curve (MLP) plot is strongly concave and rises steeply toward high true positive rate at low false positive rate, consistent with the reported AUC 0.995.





Although the second approach significantly improved methodological rigor by enforcing strict train/test separation and ensuring that all models were evaluated on identical splits, the feature set (DEG genes) was still derived from Phase 7 exploratory analysis. Since this gene selection step was originally performed using the full dataset, it may still introduce a residual form of information leakage, even though it is weaker than in the first approach. Therefore, the high AUC observed for the neural network should not be interpreted as fully unbiased performance.

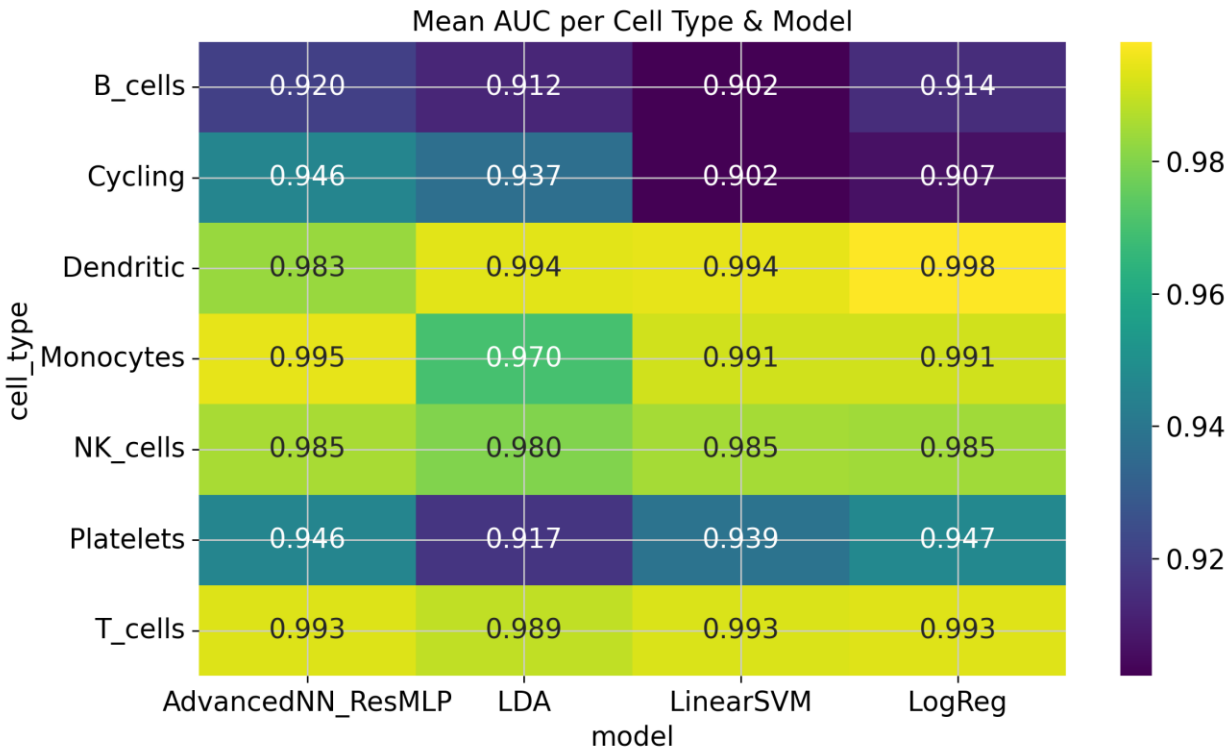
3.3 Comparison of Models:

In this section, we evaluate the performance of the four implemented classifiers—Linear Discriminant Analysis (LDA), Logistic Regression (LogReg), Linear Support Vector Machine (Linear SVM), and the Advanced Neural Network (ResMLP-based)—utilizing the Leave-One-Severity-Out (LOSO) validation framework. The comparison is structured at two levels: cell-type-specific performance and severity-level generalization.

- **Performance Evaluation across Cell Types:**

The analysis of the "Mean AUC per Cell Type & Model" heatmap reveals that all models achieved exceptional discriminative power, with most AUC values exceeding 0.90.

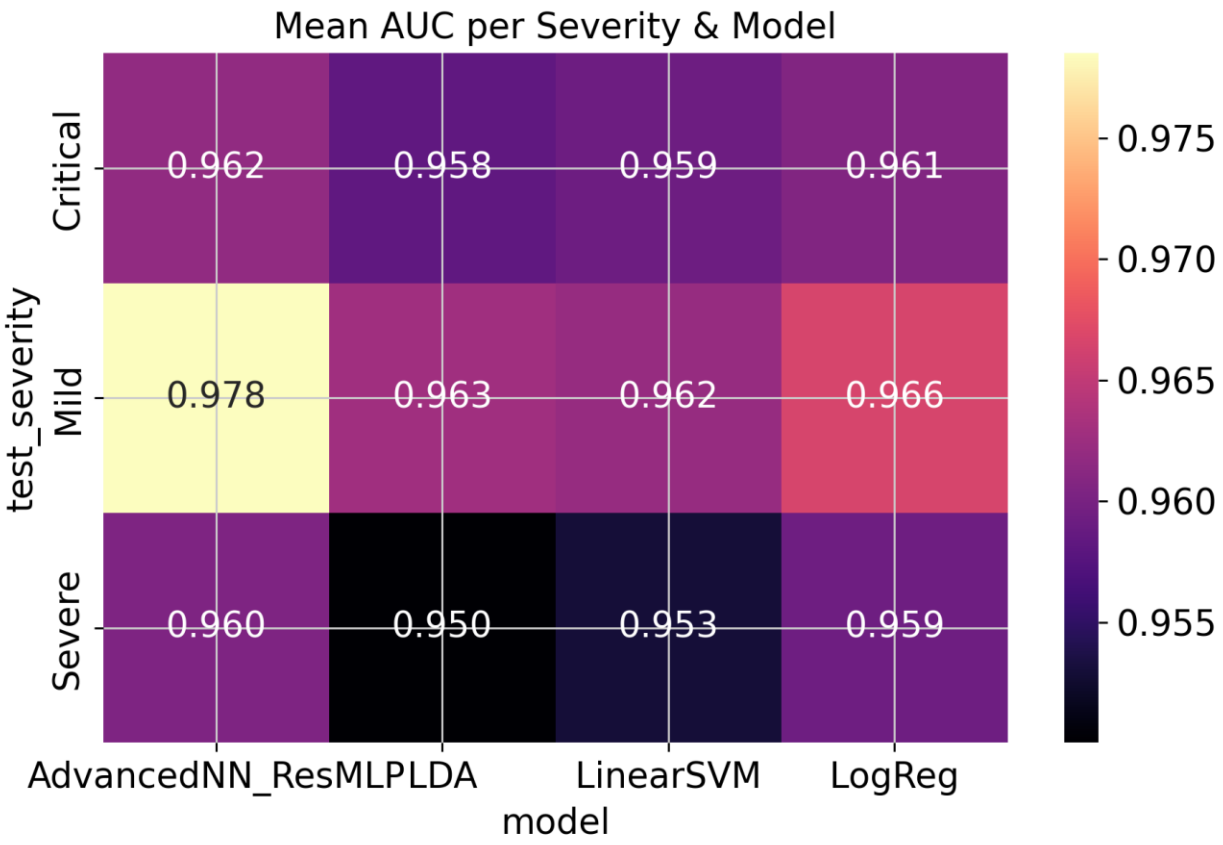
- High-Performing Lineages: For immune populations such as T cells, NK cells, Monocytes, and Dendritic cells, performance reached near-perfect levels (AUC ≈ 0.98–0.99). In Dendritic cells, Logistic Regression achieved the highest AUC (0.998), suggesting that the disease-related transcriptional shifts in these cells are highly distinct and linearly separable.
- Complex Populations: In Cycling cells, the Advanced Neural Network (ResMLP) showed a notable advantage (AUC 0.946) compared to Linear SVM (0.902) and LogReg (0.907). This indicates that cells in active proliferation may harbor non-linear transcriptomic structures that benefit from the deeper architecture of a neural network.
- Consistent Signatures: In NK cells and T cells, all models performed nearly identically (AUC ≈ 0.98–0.99), implying a robust and consistent biological signal across the lymphoid lineage that is easily captured by even the simplest linear models.



- **Generalization across Disease Severities:**

The generalization capacity of the models was tested by holding out one clinical severity in each fold. The "Mean AUC per Severity & Model" heatmap confirms the stability of the identified gene signatures.

1. Mild: Generalization to the Mild state was robust across all models, with the AdvancedNN leading at 0.978.
2. Severe: A slight decrease in performance was observed in the Severe category, where LDA showed the lowest comparative AUC (0.950), while LogReg and AdvancedNN remained stable at ~0.96.
3. Critical: All models demonstrated high and similar performance (AUC ≈ 0.96), suggesting that the transcriptomic signal distinguishing Disease from Healthy remains stable even when the Critical severity is held out.



- **Final Model Ranking and Practical Implications:**

Based on the integrated analysis of cell-type and severity-level results, the models are ranked as follows:

1. Advanced Neural Network (ResMLP): The most consistent top performer with the highest overall AUC. It showed the highest performance in Cycling cells, where linear models exhibited slightly lower AUC values.
2. Logistic Regression: Extremely competitive and often matched the neural network. It offers the best balance of stability, computational efficiency, and biological interpretability.
3. Linear SVM: Highly similar to Logistic Regression, providing robust linear boundaries.
4. LDA: Strong performance but slightly more sensitive to the feature-to-sample ratio in smaller clusters.

The observation that linear models (LogReg and SVM) perform nearly as well as deep learning architectures (with differences often <0.02 AUC) aligns perfectly with established benchmarking studies, such as PMID: 35281805. These findings reinforce that when effective feature engineering—such as our cell-type-specific DEG selection—is performed, the disease signal in scRNA-seq data becomes largely linearly separable.

4. Conclusion: Biological Interpretation and Insights

- **Biological Interpretations and Insights:** The analysis revealed that COVID-19 induces profound and cell-type-specific transcriptomic alterations that are conserved across different clinical severities. Our Differential Expression (DE) analysis identified robust gene signatures, particularly within the Myeloid (Monocytes/DCs) and NK cell populations, which consistently showed the highest discriminative power. These results suggest that the innate immune system carries a "universal signature" of the disease state that remains detectable even when testing the model on a clinical severity it has never encountered before. This conservation of transcriptomic markers underscores the potential for developing generalized diagnostic tools that do not rely on the specific phase or intensity of the illness.

- **The Optimal Predictive Model:** In evaluating the trade-off between predictive power, computational efficiency, and interpretability, Logistic Regression emerged as the most balanced and practical model for this project. While the Advanced Neural Network (ResMLP) achieved the highest overall AUC (0.9669), Logistic Regression provided superior raw accuracy (91.22%) and nearly identical AUC results. In accordance with established

literature (PMID: 35281805), our findings confirm that for transcriptomic classification, linear models are remarkably effective. They provide a transparent decision-making process—allowing researchers to trace predictions back to specific gene weights—while maintaining high performance comparable to deep learning architectures. In summary, the project successfully demonstrated that linear models (LogReg and LDA), especially when combined with effective feature selection (e.g., DEGs), can perform on par with more complex models like ANN and SVM.

5. Future Directions and Improvements

To advance this research further, several strategies are recommended:

Expansion of Biological Replicates: Incorporating additional healthy donors would mitigate the current limitation of single-donor bias and further validate the generalizability of the identified gene signatures.

Pseudo-bulk Analysis: Moving from cell-level modeling to pseudo-bulk aggregation (grouping cells by donor) could provide a more stable statistical framework for clinical-grade diagnostics.

Regularization Tuning: Exploring ElasticNet penalties within Logistic Regression could help refine feature selection by managing highly correlated gene expression patterns more effectively.

6. References

Dataset: Zheng, L., et al. (2025). *Single-cell RNA-seq of Peripheral Blood Mononuclear Cells (PBMC) from COVID-19 Patients across Different Severities*. NCBI Gene Expression Omnibus. Accession: GSE293707.

Primary Literature: He, Y., et al. (2022). *A Systematic Evaluation of Supervised Machine Learning Algorithms for Cell Phenotype Classification Using Single-Cell RNA Sequencing Data*. Frontiers in Genetics. PMID: 35281805.